Unit 7 Guided Notes — Data Analysis & Statistics

Scatterplots, correlation, the line of best fit, and two-way tables.

Standards: NC.8.SP.1 · NC.8.SP.2 · NC.8.SP.3 · NC.8.SP.4

🗐 **Blank Worksheet (PDF)** ✅ **Answer Key (PDF)** 🗐 **Official PDF Packet** ♿ **EC/Modified Notes (PDF)**

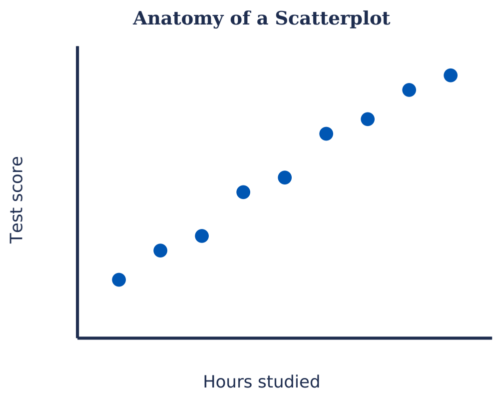
Interactive guided notes for Unit 7 — scatterplots, correlation, the line of best fit, and two-way tables. Fill in the blanks to learn, then click to check your thinking.

1 Scatterplots & Bivariate Data

📌 **NC.8.SP.1:** Construct and interpret scatter plots for bivariate measurement data.

Bivariate data has TWO values for each item — like a person's height AND weight, or hours studied AND test score. We use a scatterplot to look for patterns between the two variables.

Term	Meaning	Example
Univariate data	______	a list of test scores
Bivariate data	______	hours studied AND test score



A scatterplot has: an x-axis for the ______ variable, a y-axis for the ______ variable. Each dot represents ______ data point. Both axes need a ______ and a ______.

Try It

Suppose you collect data on 10 students: how many hours of sleep they got and their test score the next day.

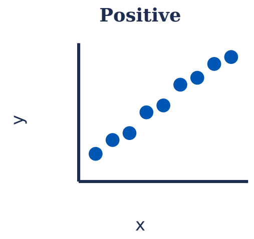
- Independent variable (x-axis)? ______
- Dependent variable (y-axis)? ______
- Total number of dots on the scatterplot: ______

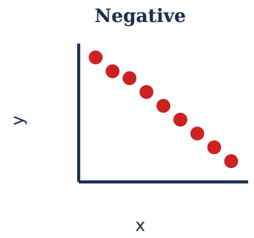
💡 **Why scatterplots?** Bar graphs show frequencies of categories. Scatterplots show relationships between two numerical quantities. If you're asking "does X affect Y?", you want a scatterplot.

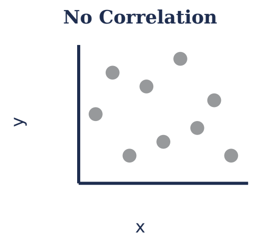
2 Correlation

📌 **NC.8.SP.1:** Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.

Correlation describes how two variables tend to change together — **positive**, **negative**, or **none**. Plus two strengths: **strong** or **weak**.

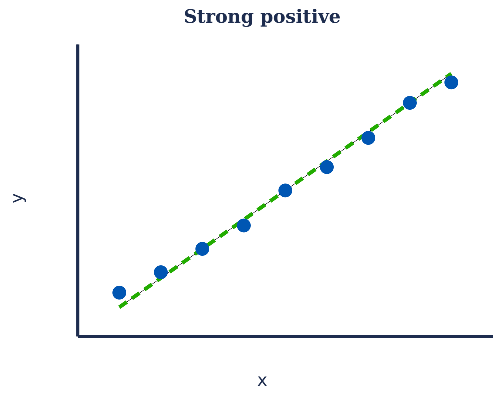


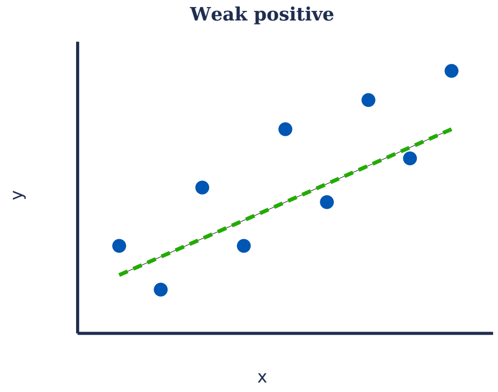




Positive correlation: as x increases, y also ______. Negative correlation: as x increases, y ______. No correlation: the dots are scattered with ______ clear pattern.

Strength of Correlation





Strong = dots tightly hug the trend line. Weak = dots are spread out around the trend.

Try It — Predict the correlation type

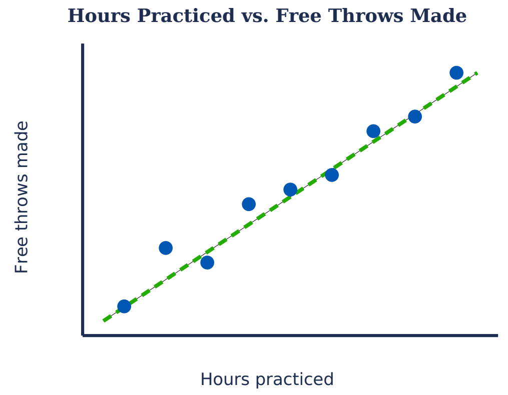
- Hours studied vs. test score → ______
- Number of absences vs. final grade → ______
- Shoe size vs. IQ → ______
- Outdoor temperature vs. ice cream sales → ______
- Age of a car vs. value of the car → ______

⚠ **CORRELATION ≠ CAUSATION:** Just because two things are correlated doesn't mean one causes the other! Ice cream sales and shark attacks are both higher in summer — but ice cream doesn't cause sharks. Both are caused by hot weather.

3 Line of Best Fit

📌 **NC.8.SP.2 / NC.8.SP.3:** Informally fit a straight line for scatter plots that suggest a linear association; use the equation to solve problems and make predictions.

When the dots suggest a linear pattern, we can draw a single straight line through the "middle" of the cloud of dots. This line represents the overall trend and helps us make predictions.



Notice the line: passes through the "middle" of the dots · has roughly the same number of dots above and below it · follows the overall positive trend · doesn't have to touch any specific dot.

Drawing a Line of Best Fit (Informally)

1. Look at the overall direction of the dots (positive or negative).
2. Use a ruler — put the line so dots are roughly balanced above and below.
3. Don't connect the dots! It's a smooth straight line, not a zigzag.
4. Don't be tricked by outliers — ignore dots far from the rest.

Try It — $y = 2x + 1$ (x = hours studied, y = test score)

- The slope means: each extra hour of study adds about ______ points.
- The y-intercept means: with 0 hours of study, the predicted score is ______.
- If a student studies 4 hours, predicted score = $2(4) + 1$ = ______.
- If a student studies 8 hours, predicted score = ______.

⚠ **Predictions are approximate.** The line is a model, not exact. A student who studies 4 hours might get 7 or 11 — the line predicts about 9 on average.

4 Two-Way Tables

📌 **NC.8.SP.4:** Understand patterns of association between two variables by displaying frequencies and relative frequencies in a two-way table.

Two-way tables organize categorical bivariate data — when both variables are categories instead of numbers. Example: grade level and lunch preference.

Reading a Two-Way Table — 120 students surveyed about pizza

	Like Pizza	Don't Like Pizza	Total
6th Grade	32	8	40
7th Grade	28	12	40
8th Grade	24	16	40
Total	**84**	**36**	**120**

- Total students surveyed: ______
- Number of 6th graders who like pizza: ______
- Total 7th graders surveyed: ______
- Total students who don't like pizza: ______

Relative Frequency

Relative frequency = the count divided by a total.

- Fraction of all 120 students who like pizza: 84 ÷ 120 = ______ or ______
- Fraction of 6th graders who like pizza: 32 ÷ 40 = ______ or ______
- Fraction of 8th graders who like pizza: 24 ÷ 40 = ______ or ______

💡 **Looking for an association:** 80% of 6th graders like pizza but only 60% of 8th graders do. There's an association between grade level and pizza preference — they're related!

Try It — 100 students surveyed about pets

	Has pet	No pet	Total
Boys	20	30	50
Girls	35	15	50
Total	**55**	**45**	**100**

- Fraction of boys who have a pet: 20 ÷ 50 = ______ or ______
- Fraction of girls who have a pet: 35 ÷ 50 = ______ or ______
- Is there an association between gender and pet ownership? ______

Error Analysis

SPOT THE ERROR **"Ice cream sales and drowning rates have a positive correlation, so eating ice cream causes drowning."**

Correlation does NOT mean causation! Both increase in summer because of a lurking variable — hot weather. Always look for a third variable that could explain the relationship.

SPOT THE ERROR **"The line of best fit must pass through every data point."**

The line of best fit shows the general trend — it should have roughly equal points above and below, but does NOT need to pass through any specific point.

SPOT THE ERROR **"More 8th graders like basketball (35) than 7th graders (30), so 8th graders are more into basketball." (8th grade has 80 students total vs 70 in 7th grade!)**

Use relative frequency to compare fairly: 7th = 30/70 ≈ 42.9% vs 8th = 35/80 ≈ 43.75%. They're nearly the same! When comparing groups of different sizes, ALWAYS use percentages, not raw counts.

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